

# Crawling the Web with Scrapy

Web crawling or spidering is the process of systematically extracting data from a website using a Web crawler, spider or robot. A Web scraper methodically harvests data from a website. This article takes the reader through the Web scraping process using Scrapy.



Scrapy is one of the most powerful and popular Python frameworks for crawling websites and extracting structured data useful for applications like data analysis, historical archival, knowledge processing, etc.

To work with Scrapy, you need to have Python installed on your system. Python can be downloaded from [www.python.org](http://www.python.org).

## Installing Scrapy with Pip

Pip is installed along with Python in the `Python/Scripts/folder`. To install Scrapy, type the following command:

```
pip install scrapy
```

The above command will install Scrapy on your machine in the `Python/Lib/site-packages` folder.

## Creating a project

With Scrapy installed, navigate to the folder in which you want to create your project, open cmd and type the command below to create the Scrapy project:

```
scrapy startproject scrapy_first
```

The above command will create a Scrapy project with the

following file structure:

```
"scrapy_first/
  - scrapy.cfg
  scrapy_first/
    - __init__.py
    - items.py
    - pipelines.py
    - settings.py
    - spiders/
      - __init__.py"
```

In the folder structure given above, 'scrapy\_first' is the root directory of our Scrapy project.

A spider is a class that describes how a website will be scraped, how it will be crawled and how data will be extracted from it. The customisation needed to crawl and parse Web pages is defined in the spiders.

## A spiders's scraping life cycle

1. You start by generating the initial request to crawl the first URL obtained by the `start_requests()` method, which generates a request for the URLs specified in the `start_urls`, and parses them using the `parse` method as a